

Long-Term High Flow Nasal Cannula Therapy in Primary and Secondary Bronchiectasis.

PMID: 37174791 | Indexed in PubMed

Bronchiectasis is the consequence of chronic bronchial inflammation, inappropriate mucus clearance, bacterial colonization, and recurrent or chronic infection. High flow therapy (HFT) is a type of non-invasive respiratory therapy, usually delivered through a nasal cannula interface (HFNC). It delivers heated and humidified air with a stable fraction of inspired oxygen and a wide range of possible flow rates. Determine the effectiveness of HFNC as add-on therapy in adult primary and secondary bronchiectasis with frequent acute exacerbations (AEs) and/or hospitalizations. This is a single-center crossover study on long-term home therapy with HFNC in adult bronchiectasis. Pharmacological therapy included pulse therapy with mucolytics and bronchodilators. After one year, all patients were switched to additional HFNC. The temperature range was 31-37 °C. The flow range was 35-60 L/m. FiO₂ was 0.21. Seventy-eight patients completed the follow-up; 54% were females; the median age was 70 years (IQR 60-76). The etiology of bronchiectasis was mainly post-infective (51%), COPD related (26%), and congenital (11%). AEs at baseline were 2.81 (± 2.15). A significant reduction in AEs was observed after 24 months with a mean of 0.45 (± 0.66) (f-ratio value 79.703. p-value < 0.00001). No significant difference was observed after HFNC therapy on FEV₁ (2.39 ± 0.87 vs. 2.55 ± 0.82 ; f-ratio 0.79. p-value 0.45) and FVC (2.73 ± 0.88 vs. 2.84 ± 0.90 ; f-ratio 0.411. p-value 0.66). A significant reduction in mMRC score was observed after HFNC therapy (2.40 ± 0.81 vs. 0.97 ± 0.97 at 2 months vs. 0.60 ± 0.78 at 24 months; f-ratio value 95.512. p-value < 0.00001). HFNC is a well-tolerated add-on therapy for adult bronchiectasis. Dyspnea improved after 2 months and further after 2 years. The exacerbation rate decreased during the 2 years follow-up. No significant difference was observed in lung function.

Utility of a personalised Bronchiectasis Action Management Plan (BAMP) for children with bronchiectasis: protocol for a multicentre, double-blind parallel, superiority randomised controlled trial.

PMID: 34937712 | Indexed in PubMed

Bronchiectasis is no longer considered rare or irreversible in children, yet it remains relatively under-researched and neglected in respiratory health globally. Bronchiectasis (including chronic suppurative lung disease) causes substantial morbidity for patients and significant impact on caregivers, especially during acute respiratory exacerbations. In other chronic respiratory diseases (eg, asthma), empowering consumers with an individualised plan for management of acute exacerbations improves clinical outcomes. However, in the absence of any such data specific to bronchiectasis, action management plans are rarely currently used in children or adults with bronchiectasis. We hypothesise that providing an individualised bronchiectasis action management plan (BAMP) to children with bronchiectasis reduces non-scheduled doctor consultations, compared with not having a BAMP. This multicentre, parallel, double-blind, randomised trial involving three urban Australian hospitals commenced in June 2018 and will include 198 children, aged <19 years with bronchiectasis who had 2 or more exacerbations in the previous 18 months. Children will be randomised to having an individualised BAMP or standard care (a decoy clinic letter). Primary caregivers will then be followed up monthly for 12 months. The primary outcome is the rate of acute non-scheduled doctor visits for respiratory exacerbations by 12 months. The main secondary outcomes are cough-specific quality of life scores at 6 and 12 months, overall exacerbation rate over 12 months, and proportion of children who received timely influenza vaccination by 30 May annually. The Human Research Ethics Committees of the Northern

Territory Department of Health and Menzies School of Health Research and Queensland Children's Hospital approved the study. The results of the trial will be submitted for publication and the BAMP made available free online. Australia and New Zealand Clinical Trials Register ACTRN12618000604202.

[Annual review of bronchiectasis in 2021].

PMID: 35658383 | Indexed in PubMed

Bronchiectasis, a common chronic respiratory disease, is characterized by irreversible and abnormal bronchial dilatation due to multiple causes. In 2021, a series of significant research progress have been made in bronchiectasis, focusing on the mechanism, diagnosis, clinical phenotypes, treatment, comorbidities, etc. Several studies have shown that many mediators are involved in the pathogenesis of bronchiectasis, such as lipids, platelets, and respiratory microorganisms, providing new insights into the development of prevention and therapy targets of bronchiectasis. At the same time, an international expert consensus proposed radiological and clinical diagnosis criteria for inclusion of bronchiectasis patients in clinical trials. Moreover, the proposal of new phenotype, measurement tools and predictors on bronchiectasis, promote the process of individualized therapy for patients. Regarding the clinical trials on bronchiectasis, several important studies have been published, including tobramycin inhalation powder, airway clearance techniques, tiotropium bromide and so forth. In addition, researches on bronchiectasis comorbidities also have new findings. This review summarized the recent published literatures in order to help clinicians better understand bronchiectasis.

Clinical characteristics and etiologies of bronchiectasis in Korean children: A multicenter retrospective study.

PMID: 30961955 | Indexed in PubMed

Bronchiectasis is a chronic pulmonary disease characterized by progressive and irreversible bronchial dilatation. The aim of the present study was to investigate the etiologies and clinical features of bronchiectasis in Korean children. We performed a retrospective review of the medical records for children diagnosed with bronchiectasis between 2000 and 2017 at 28 secondary or tertiary hospitals in South Korea. A total of 387 cases were enrolled. The mean age at diagnosis was 9.2 ± 5.1 years and 53.5% of the patients were boys. The most common underlying cause of bronchiectasis was preexisting respiratory infection (55.3%), post-infectious bronchiolitis obliterans (14.3%), pulmonary tuberculosis (12.3%), and heart diseases (5.6%). Common initial presenting symptoms included chronic cough (68.0%), recurrent pneumonia (36.4%), fever (31.1%), and dyspnea (19.7%). The most predominantly involved lesions were left lower lobe (53.9%), right lower lobe (47.1%) and right middle lobe (40.2%). No significant difference was observed in the distribution of these involved lesions by etiology. The forced expiratory volume in 1 s (FEV1) levels were lowest in cases with interstitial lung disease-associated bronchiectasis, followed by those with recurrent aspiration and primary immunodeficiency. Bronchiectasis should be strongly considered in children with chronic cough and recurrent pneumonia. Long-term follow-up studies on pediatric bronchiectasis are needed to further clarify the prognosis and reduce the disease burden in these patients.

X-linked inheritance of primary ciliary dyskinesia and retinitis pigmentosa due to RPGR variant: A case report and literature review.

PMID: 37915370 | Indexed in PubMed

Bronchiectasis is a chronic respiratory condition characterized by irreversible bronchial dilation, often caused by infection or inflammation. It can be associated with primary ciliary dyskinesia (PCD), a hereditary disorder affecting cilia function in various organs and flagella. PCD's genetic heterogeneity leads to varying disease severity. PCD may be more prevalent in Asia, but its diagnosis is often delayed in Japan. This study reviewed a case of PCD and retinitis pigmentosa (RP) with the relevant literature. The patient had a persistent cough, sputum, and diffuse bronchiectasis. He was diagnosed with a combination of PCD and RP, with the presence of an X-linked retinitis pigmentosa GTPase regulator (RPGR) variant confirmed through electron microscopy, retinal scan, and genetic testing. Although co-occurrence of bronchiectasis and RP is rare, PCD should be considered in cases of persistent wet cough in childhood or unidentified bronchiectasis aetiology. Ophthalmologists should consider concomitant PCD in RP patients.

[Management of adult bronchiectasis - Consensus-based Guidelines for the German Respiratory Society (DGP) e.V. (AWMF registration number 020-030)].

PMID: 39515342 | Indexed in PubMed

Bronchiectasis is an etiologically heterogeneous, chronic, and often progressive respiratory disease characterized by irreversible bronchial dilation. It is frequently associated with significant symptom burden, multiple complications, and reduced quality of life. For several years, there has been a marked global increase in the prevalence of bronchiectasis, which is linked to a substantial economic burden on healthcare systems. This consensus-based guideline is the first German-language guideline addressing the management of bronchiectasis in adults. The guideline emphasizes the importance of thoracic imaging using CT for diagnosis and differentiation of bronchiectasis and highlights the significance of etiology in determining treatment approaches. Both non-drug and drug treatments are comprehensively covered. Non-pharmacological measures include smoking cessation, physiotherapy, physical training, rehabilitation, non-invasive ventilation, thoracic surgery, and lung transplantation. Pharmacological treatments focus on the long-term use of mucolytics, bronchodilators, anti-inflammatory medications, and antibiotics. Additionally, the guideline covers the challenges and strategies for managing upper airway involvement, comorbidities, and exacerbations, as well as socio-medical aspects and disability rights. The importance of patient education and self-management is also emphasized. Finally, the guideline addresses special life stages such as transition, family planning, pregnancy and parenthood, and palliative care. The aim is to ensure comprehensive, consensus-based, and patient-centered care, taking into account individual risks and needs.

Infection prevention and chronic disease management in cystic fibrosis and noncystic fibrosis bronchiectasis.

PMID: 32160809 | Indexed in PubMed

Bronchiectasis is a chronic lung disease (CLD) characterized by irreversible bronchial dilatation noted on computed tomography associated with chronic cough, ongoing viscid sputum production, and recurrent

pulmonary infections. Patients with bronchiectasis can be classified into two groups: those with cystic fibrosis and those without cystic fibrosis. Individuals with either cystic fibrosis related bronchiectasis (CFRB) or noncystic fibrosis related bronchiectasis (NCFRB) experience continuous airway inflammation and suffer airway architectural changes that foster the acquisition of a unique polymicrobial community. The presence of microorganisms increases airway inflammation, triggers pulmonary exacerbations (PEx), reduces quality of life (QOL), and, in some cases, is an independent risk factor for increased mortality. As there is no cure for either condition, prevention and control of infection is paramount. Such an undertaking incorporates patient/family and healthcare team education, immunoprophylaxis, microorganism source control, antimicrobial chemoprophylaxis, organism eradication, daily pulmonary disease management, and, in some cases, thoracic surgery. This review is a summary of recommendations aimed to thwart patient acquisition of pathologic organisms, and those therapies known to mitigate the effects of chronic airway infection. A thorough discussion of airway clearance techniques and treatment of or screening for nontuberculous mycobacteria (NTM) is beyond the scope of this discussion.

[Bronchiectasis not related to cystic fibrosis in children: Guidelines for diagnosis, monitoring and treatment].

PMID: 33231063 | Indexed in PubMed

Bronchiectasis is within the spectrum of chronic lung disease characterized by progressive and often irreversible bronchial dilation caused by structural changes in the bronchial wall and chronic inflammation of the airways. The cardinal symptom is persistent moist and productive chronic cough that should alert to timely intervention and interrupt the cycle of inflammation, infection, and airway damage. Early diagnosis through high-resolution computed tomography of the chest and clinical monitoring facilitate the implementation of intensive treatment that reduces and minimizes damage to the airway. Although current therapeutic actions for the management of bronchiectasis are effective, there are few randomized clinical trials in pediatrics. The objective of the document is to provide an update on the diagnosis, monitoring and treatment of bronchiectasis not related to cystic fibrosis in children.

Pediatric bronchiectasis: No longer an orphan disease.

PMID: 26840008 | Indexed in PubMed

Bronchiectasis is described classically as a chronic pulmonary disorder characterized by a persistent productive cough and irreversible dilatation of one or more bronchi. However, in children unable to expectorate, cough may instead be wet and intermittent and bronchial dilatation reversible in the early stages. Although still considered an orphan disease, it is being recognized increasingly as causing significant morbidity and mortality in children and adults in both affluent and developing countries. While bronchiectasis has multiple etiologies, the final common pathway involves a complex interplay between the host, respiratory pathogens and environmental factors. These interactions lead to a vicious cycle of repeated infections, airway inflammation and tissue remodelling resulting in impaired airway clearance, destruction of structural elements within the bronchial wall causing them to become dilated and small airway obstruction. In this review, the current knowledge of the epidemiology, pathobiology, clinical features, and management of bronchiectasis in children are summarized. Recent evidence has emerged

to improve our understanding of this heterogeneous disease including the role of viruses, and how antibiotics, novel drugs, antiviral agents, and vaccines might be used. Importantly, the management is no longer dependent upon extrapolating from the cystic fibrosis experience. Nevertheless, substantial information gaps remain in determining the underlying disease mechanisms that initiate and sustain the pathophysiological pathways leading to bronchiectasis. National and international collaborations, standardizing definitions of clinical and research end points, and exploring novel primary prevention strategies are needed if further progress is to be made in understanding, treating and even preventing this often life-limiting disease.

Diagnostic imaging in adult non-cystic fibrosis bronchiectasis.

PMID: 31508157 | Indexed in PubMed

Radiology plays a key role in the diagnosis of bronchiectasis, defined as permanent dilatation of the bronchial lumen. Volumetric thin-section multidetector computed tomography is an excellent noninvasive modality to evaluate bronchiectasis. Bronchiectasis is categorised by morphological appearance. Cylindrical bronchiectasis has a smooth tubular configuration and is the most common form. Varicose bronchiectasis has irregular contours with alternating dilating and contracting lumen. Cystic bronchiectasis is the most severe form and exhibits saccular dilatation of bronchi. Bronchial dilatation is the hallmark of bronchiectasis and is evaluated in relation to the accompanying pulmonary artery. A broncho-arterial ratio exceeding 1:1 should be considered abnormal. Normal bronchi are narrower in diameter the further they are from the lung hila. Lack of normal bronchial tapering over 2 cm in length, distal from an airway bifurcation, is the most sensitive sign of bronchiectasis. Findings commonly associated with bronchiectasis include bronchial wall thickening, mucus plugging and tree-in-bud opacities. Bronchiectasis results from a myriad of conditions, with post-infectious bronchiectasis being the most common. Imaging can sometimes discern the cause of bronchiectasis. However, in most cases it is nonspecific or only suggestive of aetiology. While morphological types are nonspecific, the distribution of abnormality offers clues to aetiology. Bronchiectasis is a chronic progressive condition with significant disease burden and frequent exacerbations, for which the diagnosis relies on cross-sectional imaging. The major imaging findings include bronchial dilatation, bronchial contour abnormalities and visualisation of the normally invisible peripheral airways. Bronchiectasis is the end result of various conditions, including immunodeficiencies, mucociliary disorders and infections. Imaging is often nonspecific with regard to aetiology but can be suggestive. Distribution of abnormality in the lung offers helpful clues for establishing aetiology. To review the cross-sectional imaging appearance of bronchiectasis and the common associated findings. To get a sense of how radiology can aid in establishing the aetiology of bronchiectasis.